

Continuation of Exposé VII

Remark 3.8. In the case where X is noetherian and Y is regular, 3.7 completely determines the abelian group $K^\bullet(X')$ from the λ -rings $K^\bullet(X)$, $K^\bullet(Y)$, from the maps i_*, i^* between these rings, and from the element $N \in K^\bullet(Y)$. Indeed, $K^\bullet(Y')$ can be computed from $K^\bullet(Y)$ and N by (VI 1.1). We shall see in the following paragraph that the multiplicative structure of $K^\bullet(X')$ is determined by the same data, as is the λ -structure after tensoring by $\mathbb{Q}$; this restriction is probably unnecessary, cf. 4.5.

4. Regular immersions and the filtration of $K^\bullet$

Throughout this paragraph we keep the notation and hypotheses of 3.1.

Lemma 4.1. Suppose that there exists on Y a locally free sheaf N' of finite type such that, if N, N' denote the classes in $K^\bullet(Y)$ of N, N' , one has $N = N' + 2$. Then the following congruence is verified:

$$\lambda_{-1}(F) = 0 \pmod{(1 - L)}.$$

Let d be the rank of N , which may be supposed constant; then the rank of F is $d - 1$. By definition,

$$\begin{aligned} \lambda_{-1}(F) &= \sum_{i \geq 0} (-1)^i \lambda^i(F) \\ &= (-1)^{d-1} \lambda^{d-1}(F - 1) \\ &= (-1)^{d-1} \lambda^{d-1}(F + L - 1 - L) \\ &= (-1)^{d-1} \lambda^{d-1}(N - 2 + 1 - L). \end{aligned}$$

But

$$\lambda_t(1 - L) = \lambda_t(1)/\lambda_t(L) = (1 + t)/(1 + Lt).$$

Thus, for every $i \geq 1$, $\lambda^i(1 - L)$ is divisible by $1 - L$. Consequently

$$\lambda_{-1}(F) \equiv (-1)^{d-1} \lambda^{d-1}(N - 2) \equiv (-1)^{d-1} \lambda^{d-1}(N') \pmod{(1 - L)}.$$

Since N' has rank $d - 2$, $\lambda^{d-1}(N') = 0$, which gives the result.

4.2. Let $i : Y \rightarrow X$ be a regular immersion. Let r be an integer ≥ 1 , and let

$$s : X \longrightarrow \mathbf{P}_X^r$$

be the section defined by the homomorphism

$$(\mathcal{O}_X)^{r+1} \longrightarrow \mathcal{O}_X$$

obtained by quotienting by the subsheaf generated by the last r coordinates. If $\mathbf{P}_X^r$ is defined by the graded $\mathcal{O}_X$ -algebra $\mathcal{O}_X[T_0, \dots, T_r]$, the ideal defining the section s is generated by the sections $T_1, \dots, T_r$; hence I/I^2 is isomorphic to $(\mathcal{O}_X[T_0])^r$, and the corresponding ideal of $\mathcal{O}_P$ to $(\mathcal{O}_X)^r$.

Since the section s is evidently a regular immersion, the same is true of $s \circ i$, and one may write the corresponding exact sequence of conormal sheaves (1.7). We thus obtain the following equality in $K^\bullet(Y)$, denoting by N the conormal sheaf of i , and by N_1 the conormal sheaf of $s \circ i$:

$$N_1 = N + r.$$

It follows that, for every regular immersion i , one can, by composing it with a section

$$X \longrightarrow \mathbf{P}_X^2 \quad \text{or} \quad X \longrightarrow \mathbf{P}_{\mathbf{P}_X^1}^1,$$

obtain a regular immersion satisfying the property of Lemma 4.1.

Theorem 4.3. Let $i : Y \rightarrow X$ be a regular closed immersion, and let N be the class in $K^\bullet(Y)$ of the conormal sheaf of Y in X . Suppose that X is quasi-compact and has an ample sheaf. Then, with the notation of 2.3, for every $y \in K^\bullet(Y)$ and every $p \geq 1$, one has

$$i_*(\lambda^p(N, y)) = \lambda^p(i_*(y)) \pmod{\text{torsion}}, \quad (4.3.1)$$

where $\lambda^p(N, y)$ is the element defined in (V 5.3).

First suppose that the hypothesis of 4.1 is satisfied. In this case we shall prove

$$i_*(\lambda^p(N, y)) = \lambda^p(i_*(y)).$$

Since $f_*f^* = \text{Id}$, this relation is equivalent to

$$f^*(i_*(\lambda^p(N, y))) = f^*(\lambda^p(i_*(y))).$$

But

$$\begin{aligned} f^*(\lambda^p(i_*(y))) &= \lambda^p(f^*(i_*(y))) \\ &= \lambda^p(j_*(g^*(y)\lambda_{-1}(F))), \end{aligned} \quad (3.4)$$

and, on the other hand,

$$f^*(i_*(\lambda^p(N, y))) = j_*(g^*(\lambda^p(N, y))\lambda_{-1}(F)). \quad (3.4)$$

Since $g^*(N) = F + L$, formula (V (5.4.2)) gives

$$g^*(\lambda^p(N, y))\lambda_{-1}(F) = \lambda^p(g^*(N), g^*(y))\lambda_{-1}(F) = \lambda^p(L, g^*(y))\lambda_{-1}(F).$$

Putting $z = g^*(y)\lambda_{-1}(F)$, we are therefore reduced to proving

$$j_*(\lambda^p(L, z)) = \lambda^p(j_*(z)),$$

that is, the analogous relation for the immersion j and the element z . By hypothesis, $\lambda_{-1}(F)$ is a multiple of $1 - L$; hence one can find z' such that $z = z'(1 - L)$. Then $z = j^*(u')$, with $u' = j_*(z')$ by (2.7). Since $L = j^*(J')$, one finally has to verify

$$j_*(j^*(\lambda^p(J', u'))) = \lambda^p(j_*(j^*(u'))).$$

The exact sequence

$$0 \longrightarrow J' \longrightarrow \mathcal{O}_{X'} \longrightarrow \mathcal{O}_{Y'} \longrightarrow 0$$

gives the relation

$$j_*(1) = 1 - J'. \quad (4.3.2)$$

Thus the relation to be verified is finally written, applying the projection formula,

$$\lambda^p(J', u')(1 - J') = \lambda^p(u'(1 - J')).$$

Since $1 - J' = \lambda_{-1}(J')$, this is the formula which serves to define $\lambda^p(J', u')$.

Now pass to the general case. Compose i with a section $X \rightarrow \mathbf{P}_X^1$, and then with a section $\mathbf{P}_X^1 \rightarrow \mathbf{P}_{\mathbf{P}_X^1}^1$, in such a way that the composite immersion satisfies the condition of 4.1 (cf. 4.2). Proposition 4.3 is then true for the composite immersion by what precedes; it remains to descend to X . One descends first to $\mathbf{P}_X^1$, and then to X , by applying twice the following lemma, due to J.-P. Jouanolou.

Lemma. Under the conditions of 4.3, let $s : X \rightarrow \mathbf{P}_X^1$ be the section defined in 4.2. If Proposition 4.3 is true for $s \circ i$, then it is true for i .

If 4.3 is assumed true for $s \circ i$, then, for every $p \geq 1$ and every $y \in K^\bullet(Y)$, one has

$$(s \circ i)_*(\lambda^p(N + 1, y)) = \lambda^p((s \circ i)_*(y)) \pmod{\text{torsion}}. \quad (4.3.3)$$

The section s is defined by an ideal isomorphic to $\mathcal{O}_P(-1)$ by 4.2; let ξ denote its class in $K^\bullet(\mathbf{P}_X^1)$. Then, for every $x \in K^\bullet(X)$, one has

$$s_*(x) = p^*(x)(1 - \xi),$$

where $p : \mathbf{P}_X^1 \rightarrow X$ is the structural morphism. Indeed, since $p \circ s = \text{Id}_X$, one has $x = s^*(p^*(x))$, whence

$$s_*(x) = s_*(s^*(p^*(x))) = p^*(x)s_*(1),$$

and since $s_*(1) = 1 - \xi$, the desired relation follows.

Thus (4.3.3) becomes

$$\begin{aligned} (s \circ i)_*(\lambda^p(N + 1, y)) &= \lambda^p(p^*(i_*(y))(1 - \xi)) && \pmod{\text{torsion}} \\ &= \lambda^p(\xi, p^*(i_*(y)))(1 - \xi) && \pmod{\text{torsion}}. \end{aligned}$$

But

$$\lambda^p(\xi, p^*(i_*(y))) - \lambda^p(1, p^*(i_*(y)))$$

is divisible by $1 - \xi$. To see this, one may work in the universal λ -ring used to define $\lambda^p(\xi, X)$ (V 5): then $\lambda^p(\xi, X) - \lambda^p(1, X)$ is a polynomial in ξ , zero for $\xi = 1$, hence divisible by $\xi - 1$. Moreover, the equation of dependence of ξ over $K^\bullet(X)$ is $(1 - \xi)^2 = 0$ (VI (1.13.1)). Therefore

$$\begin{aligned} (s \circ i)_*(\lambda^p(N + 1, y)) &= \lambda^p(1, p^*(i_*(y)))(1 - \xi) && \pmod{\text{torsion}} \\ &= p^*(\lambda^p(1, i_*(y)))(1 - \xi) && \pmod{\text{torsion}}. \end{aligned}$$

Applying p_* , one obtains

$$i_*(\lambda^p(N + 1, y)) = \lambda^p(1, i_*(y)) \pmod{\text{torsion}}, \quad (4.3.4)$$

because $p_*(\xi) = 0$.

By formula (V 5.6), applied to $L = 1$, $\lambda^p(N + 1, y)$ is the coefficient of t^p in

$$\frac{\sum_{k \geq 1} \lambda^k(N, y)t^k}{1 + \sum_{k \geq 1} \lambda^k(N, y)\lambda_{-1}(N)t^k}.$$

Likewise, the same formula applied to $N = 0$ and $L = 1$ shows that $\lambda^p(1, y)$ is the coefficient of t^p in

$$\frac{\sum_{k \geq 1} \lambda^k(y)t^k}{\sum_{k \geq 0} \lambda^k(y)t^k}.$$

The congruences (4.3.4) may therefore be written

$$i_* \left(\frac{\sum_{k \geq 1} \lambda^k(N, y) t^k}{1 + \sum_{k \geq 1} \lambda^k(N, y) \lambda_{-1}(N) t^k} \right) \equiv \frac{\sum_{k \geq 1} \lambda^k(i_*(y)) t^k}{\sum_{k \geq 0} \lambda^k(i_*(y)) t^k} \pmod{\text{torsion}}.$$

Applying formula 2.8, one obtains

$$\frac{\sum_{k \geq 0} k i_*(\lambda^k(N, y)) t^k}{1 + \sum_{k \geq 1} i_*(\lambda^k(N, y)) t^k} \equiv \frac{\sum_{k \geq 0} k \lambda^k(i_*(y)) t^k}{\sum_{k \geq 0} \lambda^k(i_*(y)) t^k} \pmod{\text{torsion}}.$$

If we put

$$u(t) = 1 + \sum_{k \geq 1} i_*(\lambda^k(N, y)) t^k, \quad v(t) = \sum_{k \geq 0} \lambda^k(i_*(y)) t^k,$$

the last relation can also be written

$$\frac{d}{dt} \left(\frac{u(t)}{v(t)} \right) \equiv 0 \pmod{\text{torsion}}.$$

Since the constant term of $u(t)/v(t)$ is 1, and since all congruences are modulo torsion, one may integrate and write $u(t) \equiv v(t)$. This gives the announced congruences.

Corollary 4.4. With the notation and hypotheses of 4.3, for every $p \geq 1$ and every $y \in K^\bullet(Y)$, one has

$$i_*(\gamma^p(N, y)) = \gamma^p(i_*(y)) \pmod{\text{torsion}}.$$

This follows immediately from 4.3, using the expression of the γ^p 's as integral linear combinations of the λ^p 's.

Remark 4.5. The proof itself of 4.3 suggests that (4.3.1) should be true not only modulo torsion but also as an equality. In RRR, the equality was proved in characteristic 0 by means of a geometric characterization of $\lambda^p(N, x)$. An extension of this method to all characteristics would probably be possible if one knew how to define exterior-power operations on perfect complexes (cf. XIV).

Theorem 4.6. Let X be a quasi-compact scheme with an ample sheaf, let $d > 0$ be an integer, and let $i : Y \rightarrow X$ be a regular immersion of codimension d . Then, for every k ,

$$i_*((\text{Fil}^k K^\bullet(Y)) \otimes_{\mathbb{Z}} \mathbb{Q}) \subset (\text{Fil}^{k+d} K^\bullet(X)) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

First observe that $i_*(K^\bullet(Y)) \subset \text{Fil}^1 K^\bullet(X)$. Indeed, $X - Y$ is dense in X ; otherwise, Y would contain an affine open subset, and on that open subset the ideal J defining Y would be a nilideal, hence a locally nilpotent ideal because it is of finite type. This is impossible, since the J^n/J^{n+1} are locally free $\mathcal{O}_Y$ -modules of nonzero rank. The augmentation in $K^\bullet(X)$ of an element of $i_*(K^\bullet(Y))$ is a locally constant function on X ; since it is zero on $X - Y$, it is zero on all of X .

For every abelian group A , put $A_{\mathbb{Q}} = A \otimes_{\mathbb{Z}} \mathbb{Q}$. Let $x \in \text{Fil}^k K^\bullet(Y)_{\mathbb{Q}}$. One may write x in the form

$$x = \sum a \gamma^{i_1}(x_1) \cdots \gamma^{i_n}(x_n),$$

with $\varepsilon(x_i) = 0$ for every i , $a \in \mathbb{Q}$, and $i_1 + \cdots + i_n \geq k$. The x_i may be expressed as integral linear combinations of the classes E_α of finitely many locally free $\mathcal{O}_Y$ -modules E_α of finite type. Let R be the $\mathbb{Q}$ - λ -subalgebra of $K^\bullet(Y)_{\mathbb{Q}}$ generated by the E_α 's and by N . Then

$$x \in \text{Fil}^k R.$$

Moreover, the λ -ring filtration of R is discrete (VI 6.2); let m be such that $\text{Fil}^m R = 0$. Finally, after decomposing X into a finite sum of disjoint open subsets, and proving the assertion on

each of them (since $K^\bullet$ commutes with finite direct sums), one may suppose the E_α 's of constant rank r_α . The augmentation ring of R is then $\mathbb{Q}$.

We shall show that, for all i ,

$$i_*(\mathrm{Fil}^i R) \subset \mathrm{Fil}^{i+d} K^\bullet(X)_{\mathbb{Q}}.$$

This will imply in particular

$$i_*(x) \in \mathrm{Fil}^{k+d} K^\bullet(X),$$

and so will prove the theorem.

The proof is by descending induction on i ; the case $i = m$ is trivial. Let $y \in \mathrm{Fil}^i R$, and put, for every j ,

$$a_j = (-1)^{j-1}(j-1)!.$$

By V 6.10,

$$\gamma^{d+i}(N, y) - a_{d+i}y \in \mathrm{Fil}^{i+1} R.$$

Thus the induction hypothesis may be applied, assuming

$$i_*(\mathrm{Fil}^{i+1} R) \subset \mathrm{Fil}^{i+d+1} K^\bullet(X)_{\mathbb{Q}}.$$

Consequently

$$i_*(\gamma^{d+i}(N, y) - a_{d+i}y) \in \mathrm{Fil}^{i+d+1} K^\bullet(X)_{\mathbb{Q}}.$$

By 4.4 this entails

$$\gamma^{d+i}(i_*(y)) - a_{d+i}i_*(y) \in \mathrm{Fil}^{i+d+1} K^\bullet(X)_{\mathbb{Q}}.$$

Since $i_*(y)$ has augmentation zero,

$$\gamma^{d+i}(i_*(y)) \in \mathrm{Fil}^{d+i} K^\bullet(X).$$

As the calculations are made modulo torsion, we find

$$i_*(y) \in \mathrm{Fil}^{i+d} K^\bullet(X)_{\mathbb{Q}}.$$

Remark 4.6.1. One sees that, even if 4.3 were true without tensoring by $\mathbb{Q}$, the preceding proof would work only modulo torsion. For a counterexample, see XIV 4.6.

4.7. We return to the situation of 3.7, where we had the commutative diagram

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ g \downarrow & & \downarrow f \\ Y & \xrightarrow{i} & X. \end{array}$$

Here i is a regular immersion, X' is the X -scheme obtained by blowing up Y , and $Y' = X' \times_X Y$. Suppose that X is noetherian with an ample sheaf, and that Y is regular. We saw in 3.7 that there is then the exact sequence

$$0 \longrightarrow K^\bullet(Y) \xrightarrow{u} K^\bullet(Y') \times K^\bullet(X) \xrightarrow{v} K^\bullet(X') \longrightarrow 0.$$

We shall show that this exact sequence makes it possible to describe the entire situation formed by the $K^\bullet$'s of X, Y, X', Y' from the data of the λ -rings $K^\bullet(X), K^\bullet(Y)$, the maps i^* and i_* , and the element $N \in K^\bullet(Y)$, at least up to torsion and up to a new order.

Let K, K' be two λ -rings, let $i_* : K \rightarrow K'$ be an additive homomorphism, let $i^* : K' \rightarrow K$ be a λ -homomorphism, let d be an integer, and let $N \in K$ be such that $\lambda^i(N) = 0$ for $i > d$ and $\lambda^d(N)$ is invertible. Assume furthermore the following properties:

- i) i_* is K' -linear (projection formula);
- ii) i^*i_* is multiplication by $\lambda_{-1}(N)$;
- iii) for every $n \geq 1$ and every $y \in K$,

$$i_*(\lambda^n(N, y)) = \lambda^n(i_*(y)).$$

First use VI 1.1 to construct a K - λ -algebra K_1 playing the role of $K^\bullet(Y')$. Let L be an indeterminate, and let $K[L]$ be the K - λ -algebra generated by L , with L subject to the relations $\lambda^i(L) = 0$ for $i \geq 2$ (V 4.9). Put

$$P(L) = \lambda^d(N - L) = \sum_{i=0}^d (-1)^i \lambda^{d-i}(N) L^i.$$

The ideal generated by P is a λ -ideal. To see this, note first that the relations $\lambda^i(N) = 0$ for $i > d$ imply that the $\lambda^j(N - L)$, for $j \geq d$, are multiples of $\lambda^d(N - L)$. If one works in $1 + K/(P)[[t]]^+$, the image of $\lambda_t(N - L)$ is a polynomial of degree $< d$, and a universal argument then shows that its λ^i 's, for $i \geq d$, are equal to 1 (cf. V 4.8). Since in $1 + K[[t]]^+$ one has

$$\lambda_t(\lambda^d(N - L)) = \lambda^d(\lambda_t(N - L)),$$

it follows that the $\lambda^i(\lambda^d(N - L))$, for $i \geq 1$, are multiples of P . By the relations V 2.4.2, this implies that the ideal generated by P is a λ -ideal.

Put

$$K_1 = K/(P).$$

The natural homomorphism $K \rightarrow K_1$ will be denoted g^* ; it is a λ -homomorphism. Moreover K_1 is a free K -module with basis $1, L, \dots, L^{d-1}$. It follows from

$$\lambda^d(N) = \lambda^{d-1}(N - L) L$$

and from the assumed invertibility of $\lambda^d(N)$ that L is invertible. Thus K_1 also has the basis $1, L^{-1}, \dots, L^{-d+1}$ over K . Define a map

$$g_* : K_1 \longrightarrow K$$

by sending every element

$$y' = y_0 + y_1 L^{-1} + \dots + y_{d-1} L^{-d+1}$$

of K_1 to $y_0 \in K$. It is clear that the map so defined is an additive K -linear homomorphism and that $g_* g^* = \text{Id}$.

Next endow $K_1 \times K'$ with a λ -ring structure by setting

$$(y'_1, x_1) * (y'_2, x_2) = (y'_1 y'_2 (1 - L) + x_1 y'_2 + x_2 y'_1, x_1 x_2), \quad (4.7.1)$$

$$\lambda_*^n(y', x) = \left(\sum_{p=1}^n \lambda^{n-p}(x) \lambda^p(L, y'), \lambda^n(x) \right). \quad (4.7.2)$$

In these formulas, K_1 is regarded as a K' -algebra by the homomorphism $g^* i^*$. Next, for $y_1, y_2 \in K$ and $n \geq 1$, put

$$y_1 * y_2 = -y_1 y_2 \lambda_{-1}(N), \quad (4.7.3)$$

$$\lambda_*^n(y_1) = -\lambda^n(N, -y_1), \quad (4.7.4)$$

and define a homomorphism

$$u : K \longrightarrow K_1 \times K'$$

by

$$u(y) = (g^*(y)\lambda_{-1}(F), -i_*(y)), \quad F = N - L.$$

Proposition 4.7.5. The operations defined by (4.7.1) and (4.7.2) endow $K_1 \times K'$ with a λ -ring structure; and u is an injective homomorphism which identifies K , endowed with the operations defined by (4.7.3) and (4.7.4), with a λ -ideal of $K_1 \times K'$.

It is immediate to verify that $K_1 \times K'$ is a λ -ring. For example, one may refer to V 5.5, where an entirely analogous construction is made. The proof of the second part of the proposition is also carried out without difficulty, using the preceding relations, the relation

$$i_*(y_1 y_2 \lambda_{-1}(N)) = i_*(y_1) i_*(y_2),$$

which follows at once from the hypotheses, the relation $g_*(\lambda_{-1}(F)) = 1$ (cf. VI 5.9), and the relations of V 5.

Put

$$K'_1 = (K_1 \times K')/K$$

and denote by v the canonical homomorphism $K_1 \times K' \rightarrow K'_1$. We regard K'_1 as a λ -ring by the quotient structure. Define additive homomorphisms

$$j_* : K_1 \rightarrow K'_1, \quad j^* : K'_1 \rightarrow K_1, \quad f_* : K'_1 \rightarrow K', \quad f^* : K' \rightarrow K'_1.$$

The homomorphism j_* is the composite $K_1 \rightarrow K_1 \times K' \rightarrow K'_1$. Similarly, f^* is the composite $K' \rightarrow K_1 \times K' \rightarrow K'_1$. Thus

$$v = j_* + f^*.$$

Next define a homomorphism $K_1 \times K' \rightarrow K_1$ by associating to (y', x) the element

$$y'(1 - L) + g^* i^*(x);$$

this homomorphism vanishes on K , and hence defines a homomorphism

$$j^* : K'_1 \rightarrow K_1.$$

Finally define a homomorphism $K_1 \times K' \rightarrow K'$ by sending (y', x) to

$$i_* g_*(y') + x;$$

this homomorphism vanishes on K , and therefore defines a homomorphism

$$f_* : K'_1 \rightarrow K'.$$

Proposition 4.7.6.

- i) j^* and f^* are λ -homomorphisms;
- ii) j_* is K'_1 -linear and f_* is K' -linear;
- iii) $j^* j_*$ is multiplication by $1 - L$;
- iv) for every $y' \in K_1$ and every $p \geq 1$,

$$j_*(\lambda^p(L, y')) = \lambda^p(j_*(y'));$$

$$v) \quad f_* f^* = \text{Id}.$$

It is enough to run the machine.

Returning to the geometric situation recalled above, one sees that $K^\bullet(X')$ is constructed from $K^\bullet(Y)$, $K^\bullet(X)$, the maps i^*, i_* , and the element $N \in K^\bullet(Y)$ by the preceding method. The λ -operations on $K^\bullet(X')$ coincide, however, with the λ -operations defined above on K'_1 only up to torsion, so long as the relation (4.3.1) is proved only modulo torsion.

Theorem 4.8. Let $i : Y \rightarrow X$ be a regular closed immersion of constant codimension d , let X' be the scheme obtained by blowing up Y , and let $Y' = X' \times_X Y$, so that one has the commutative diagram

$$\begin{array}{ccc} Y' & \xrightarrow{j} & X' \\ g \downarrow & & \downarrow f \\ Y & \xrightarrow{i} & X. \end{array}$$

Suppose that X is quasi-compact and has an ample sheaf. Then, for every k , the following sequence is exact:

$$0 \longrightarrow \text{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \xrightarrow{u^{\text{gr}}} \text{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \times \text{Gr}^k K^\bullet(X)_\mathbb{Q} \xrightarrow{v^{\text{gr}}} \text{Gr}^k K^\bullet(X')_\mathbb{Q},$$

where, for $y \in \text{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q}$,

$$u^{\text{gr}}(y) = (g^{\text{gr}}(y)(-1)^{d-1} c^{d-1}(F), -i_{\text{gr}}(y)),$$

and, for $y' \in \text{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q}$ and $x \in \text{Gr}^k K^\bullet(X)_\mathbb{Q}$,

$$v^{\text{gr}}(y', x) = j_{\text{gr}}(y') + f^{\text{gr}}(x).$$

The notation is that of 3.1 and 3.3, and $c^{d-1}(F)$ denotes the $(d-1)$ -st Chern class of F (V 6.7). Moreover u^{gr} has a left inverse u'^{gr} , defined by

$$u'^{\text{gr}}(y', x) = g_{\text{gr}}(y').$$

If, in addition, X is noetherian and Y is regular, then the sequence

$$0 \longrightarrow \text{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \xrightarrow{u^{\text{gr}}} \text{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \times \text{Gr}^k K^\bullet(X)_\mathbb{Q} \xrightarrow{v^{\text{gr}}} \text{Gr}^k K^\bullet(X')_\mathbb{Q} \longrightarrow 0$$

is exact.

Proof. One could give, in the case where Y is regular, a purely algebraic proof in terms of the theory of λ -rings, using the description in 4.7. We shall instead calculate the proof of 3.7.

The inverse-image homomorphisms f^*, g^* , being λ -homomorphisms, are compatible with the λ -ring filtrations. They therefore define homomorphisms on the associated graded groups,

$$f^{\text{gr}} : \text{Gr}^k K^\bullet(X) \rightarrow \text{Gr}^k K^\bullet(X'), \quad g^{\text{gr}} : \text{Gr}^{k-d} K^\bullet(Y) \rightarrow \text{Gr}^{k-d} K^\bullet(Y'),$$

and the analogous homomorphisms after tensoring by $\mathbb{Q}$. Moreover, 4.6 shows that after tensoring by $\mathbb{Q}$ the homomorphisms i_*, j_* pass to the associated graded groups and therefore define

$$i_{\text{gr}} : \text{Gr}^{k-d} K^\bullet(Y)_\mathbb{Q} \rightarrow \text{Gr}^k K^\bullet(X)_\mathbb{Q}, \quad j_{\text{gr}} : \text{Gr}^{k-1} K^\bullet(Y')_\mathbb{Q} \rightarrow \text{Gr}^k K^\bullet(X')_\mathbb{Q}.$$

Finally, since

$$\lambda_{-1}(F) = (-1)^{d-1} \lambda^{d-1}(F),$$

the morphisms u^{gr} and v^{gr} are obtained by passing to the quotient from the morphisms u and v of 3.7. Hence $v^{\text{gr}}u^{\text{gr}} = 0$.

By VI 5.8, one has

$$g_*(\text{Fil}^{k-1} K^\bullet(Y')) \subset \text{Fil}^{k-d} K^\bullet(Y)$$

for every k , since $Y' = \mathbf{P}(N)$, with N of rank d . Thus one obtains a homomorphism

$$g_{\text{gr}} : \text{Gr}^{k-1} K^\bullet(Y') \rightarrow \text{Gr}^{k-d} K^\bullet(Y),$$

and an analogous homomorphism after tensoring by $\mathbb{Q}$. The homomorphism u'^{gr} is obtained from u' (3.7) by passage to the associated graded group. Since u' is a left inverse of u , u'^{gr} is a left inverse of u^{gr} , which at the same time proves the injectivity of u^{gr} .

To prove that $\text{Ker}(v^{\text{gr}}) = \mathfrak{I}(u^{\text{gr}})$, it remains to prove the following: if

$$y' \in \text{Fil}^{k-1} K^\bullet(Y')_{\mathbb{Q}}, \quad x \in \text{Fil}^k K^\bullet(X)_{\mathbb{Q}}$$

are such that

$$j_*(y') + f^*(x) \in \text{Fil}^{k+1} K^\bullet(X')_{\mathbb{Q}}, \quad (4.6.1)$$

then there exists $y \in \text{Fil}^{k-d} K^\bullet(Y)_{\mathbb{Q}}$ such that

- i) $x + i_*(y) \in \text{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}};$
- ii) $y' - g^*(y)\lambda_{-1}(F) \in \text{Fil}^k K^\bullet(Y')_{\mathbb{Q}}.$

Put

$$y = g_*(y').$$

Then by VI 5.8, $y \in \text{Fil}^k K^\bullet(Y)$. Next observe that f_* preserves the filtration; this uses the results of Exposé VIII, which causes no difficulty, since 4.8 is not used to prove that fact. Indeed, by 1.8 ii) f is a complete-intersection morphism of relative virtual dimension zero (VIII 4.2). Since f is projective, the result is a consequence of VIII 3.2. Applying f_* to (4.6.1), we obtain

$$i_*(j_*(y')) + f_*f^*(x) \in \text{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}},$$

hence, by 3.6,

$$i_*(y) + x \in \text{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}}.$$

To prove (ii), one must prove

$$y' - g^*(g_*y')\lambda_{-1}(F) \in \text{Fil}^k K^\bullet(Y')_{\mathbb{Q}}.$$

Set $z = y' - g^*(g_*y')\lambda_{-1}(F)$. Then

$$j_*(z) = j_*(y') - j_*(g^*(g_*y')\lambda_{-1}(F)) = j_*(y') - f^*(i_*y),$$

by 3.4. Relation (i) gives

$$f^*(i_*y) + f^*(x) \in \text{Fil}^{k+1} K^\bullet(X')_{\mathbb{Q}}.$$

Together with (4.6.1), this gives

$$j_*(z) \in \text{Fil}^{k+1} K^\bullet(X')_{\mathbb{Q}}.$$

Hence

$$j^*(j_*z) = z(1 - L) \in \text{Fil}^{k+1} K^\bullet(Y')_{\mathbb{Q}}.$$

Let $\bar{z}$ be the image of z in $\mathrm{Gr}^{k-1} K^\bullet(Y')_{\mathbb{Q}}$, and let $\overline{(1-L)}$ be the image of $1-L$ in $\mathrm{Gr}^1 K^\bullet(Y')_{\mathbb{Q}}$. Thus

$$\bar{z} \overline{(1-L)} = 0.$$

One may then apply VI 5.11, which remains valid after tensoring by $\mathbb{Q}$, because its proof uses only the structure of $K^\bullet(Y')$ as a free $K^\bullet(Y)$ -algebra. Hence one can find $z' \in K^\bullet(Y)_{\mathbb{Q}}$ such that

$$z - g^*(z')\lambda_{-1}(F) \in \mathrm{Fil}^k K^\bullet(Y')_{\mathbb{Q}}.$$

Applying g_* , and taking into account that $g_*z = 0$, one obtains

$$z' \in \mathrm{Fil}^{k+1-d} K^\bullet(Y)_{\mathbb{Q}}.$$

Therefore, since $\lambda_{-1}(F) \in \mathrm{Fil}^{d-1} K^\bullet(Y')_{\mathbb{Q}}$,

$$z \in \mathrm{Fil}^k K^\bullet(Y')_{\mathbb{Q}},$$

which completes the proof of exactness at the middle term.

Now suppose that X is noetherian and Y is regular, which implies that v is surjective (3.7). We saw in the proof of 3.7 that, for every $x' \in K^\bullet(X')$, there exists $y' \in K^\bullet(Y')$ such that

$$x' = f^*(f_*(x')) + j_*(y').$$

The same property is evidently still true after tensoring by $\mathbb{Q}$. Consequently, if $x' \in \mathrm{Fil}^k K^\bullet(X')_{\mathbb{Q}}$, there exists

$$(y', x) \in K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}$$

such that

$$x' = v(y', x), \quad x \in \mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

Indeed, it is enough to take $x = f_*(x')$.

Filter the ring $K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}$, defined in 4.7, by putting

$$\mathrm{Fil}^k(K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}) = \mathrm{Fil}^{k-1} K^\bullet(Y')_{\mathbb{Q}} \times \mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

It follows at once from (4.7.1) that this is a ring filtration. To prove that, for every k , the homomorphism

$$\mathrm{Fil}^k(K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}) \xrightarrow{v} \mathrm{Fil}^k K^\bullet(X')_{\mathbb{Q}}$$

is surjective, it is therefore enough to prove that, if $x' = \gamma^k(x'_1)$ with $\varepsilon(x'_1) = 0$, there exists

$$(y', x) \in \mathrm{Fil}^k(K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}})$$

such that $x' = v(y', x)$.

By the preceding paragraph, there exists

$$(y'_1, x_1) \in K^\bullet(Y')_{\mathbb{Q}} \times K^\bullet(X)_{\mathbb{Q}}$$

with $x'_1 = v(y'_1, x_1)$ and $x_1 \in \mathrm{Fil}^1 K^\bullet(X)_{\mathbb{Q}}$. By 4.7,

$$x' = v(\gamma_*^k(y'_1, x_1)).$$

But

$$\gamma_*^k(y'_1, x_1) = \left(\sum_{p=1}^k \gamma^{k-p}(x_1) \gamma^p(L, y'_1), \gamma^k(x_1) \right)$$

by immediate consequence of (4.7.2). Since x_1 has augmentation zero, $\gamma^k(x_1) \in \text{Fil}^k K^\bullet(X)_\mathbb{Q}$; and, taking V 5.7 into account,

$$\sum_{p=1}^k \gamma^{k-p}(x_1) \gamma^p(L, y'_1) \in \text{Fil}^{k-1} K^\bullet(Y')_\mathbb{Q}.$$

This gives the result.

Remark 4.9. In the case where X is noetherian and Y is regular, one can, proceeding as in 4.7, describe the graded-ring structure of $\text{Gr}^* K^\bullet(X')_\mathbb{Q}$ in terms of the graded rings $\text{Gr}^* K^\bullet(X)_\mathbb{Q}$, $\text{Gr}^{*-d} K^\bullet(Y)_\mathbb{Q}$, the additive homomorphisms i^{gr} and i_{gr} deduced from i^* and i_* , and the element $c'(N)$ of $\text{Gr}^* K^\bullet(Y)$. Likewise, from the same data one can describe the homomorphisms $f^{\text{gr}}, f_{\text{gr}}, j^{\text{gr}}, j_{\text{gr}}$ obtained from f^*, f_*, j^*, j_* by passing to the associated graded groups.

4.10. Let X be a regular scheme with an ample sheaf. It is known that the Grothendieck groups formed from the locally free finite-type $\mathcal{O}_X$ -modules and from the coherent $\mathcal{O}_X$ -modules are isomorphic. This fact is used to define on $K^\bullet(X)$ a “topological” filtration by taking $\text{Fil}_{\text{top}}^i K^\bullet(X)$ to be the subgroup generated by the classes of coherent sheaves whose support has codimension $\geq i$. It is easily shown that $\text{Fil}_{\text{top}}^i K^\bullet(X)$ is generated by the classes of the sheaves $\mathcal{O}_Z$, where Z runs through the integral closed subschemes of X of codimension $\geq i$.

Proposition 4.11. Let X be a regular scheme with an ample sheaf. Then, for every $k \geq 0$,

$$\text{Fil}_{\text{top}}^k K^\bullet(X)_\mathbb{Q} \subset \text{Fil}_\lambda^k K^\bullet(X)_\mathbb{Q}.$$

The reverse inclusion, valid without tensoring by $\mathbb{Q}$, will be proved in Exposé X.

By 4.10, it is enough to prove that, if Z is an integral closed subscheme of X of codimension k , then

$$\text{cl}(\mathcal{O}_Z) \in \text{Fil}^k K^\bullet(X)_\mathbb{Q}.$$

For this, proceed by noetherian induction on the integral closed subschemes of X . Suppose the assertion proved for the integral closed subschemes of Z , and prove it for Z . Let J be the ideal of $\mathcal{O}_X$ defining Z , and let z be the generic point of Z . Since Z is integral, J_z is the maximal ideal of $\mathcal{O}_{X,z}$, and since X is regular, J_z can be generated by an $\mathcal{O}_{X,z}$ -regular sequence. Thus one may find a sequence of sections of J , in a neighbourhood of z , forming an $\mathcal{O}_X$ -regular sequence. Consequently, there is an open neighbourhood U of z such that $Z' = Z - U$ is contained in Z , and such that $J|_U$ is a regular ideal of $\mathcal{O}_U$. The immersion

$$Z \cap U \xrightarrow{i} U$$

is then a regular immersion of codimension $\dim(\mathcal{O}_{X,z}) = \text{codim}(Z, X)$ (EGA IV 5.1.2). On the other hand, one has the exact sequence of Exposé IV

$$K^\bullet(Z') \xrightarrow{j^*} K^\bullet(X) \xrightarrow{u^*} K^\bullet(U) \longrightarrow 0,$$

where j is the immersion $Z' \rightarrow X$ and u is the immersion $U \rightarrow X$. We obtain an analogous sequence after tensoring by $\mathbb{Q}$. Now

$$u^*(\mathcal{O}_Z) = i_*(\mathcal{O}_{Z \cap U}),$$

and by 4.6 the image under u^* of the class of $\mathcal{O}_Z$ is therefore in $\text{Fil}^k K^\bullet(U)_\mathbb{Q}$. Since u^* is a surjective λ -homomorphism, it induces a surjection

$$\text{Fil}^k K^\bullet(X) \longrightarrow \text{Fil}^k K^\bullet(U).$$

Thus $u^*(1_Z) = u^*(x)$ with $x \in \mathrm{Fil}^k K^\bullet(X)$; hence there exists $y \in j_*(K^\bullet(Z'))$ such that

$$\mathrm{cl}(\mathcal{O}_Z) - y \in \mathrm{Fil}^k K^\bullet(X)_{\mathbb{Q}}.$$

But $Z' \subset Z$, and since $z \notin Z'$, $\mathrm{codim}(Z', X) > k$. Hence x can be expressed as a combination of classes of sheaves $\mathcal{O}_{Z''_\alpha}$, where the Z''_α are integral closed subschemes of Z , distinct from Z . By the induction hypothesis,

$$x \in \mathrm{Fil}^{k+1} K^\bullet(X)_{\mathbb{Q}},$$

which completes the proof.

Bibliography

- [1] Cartan-Eilenberg, *Homological Algebra*.